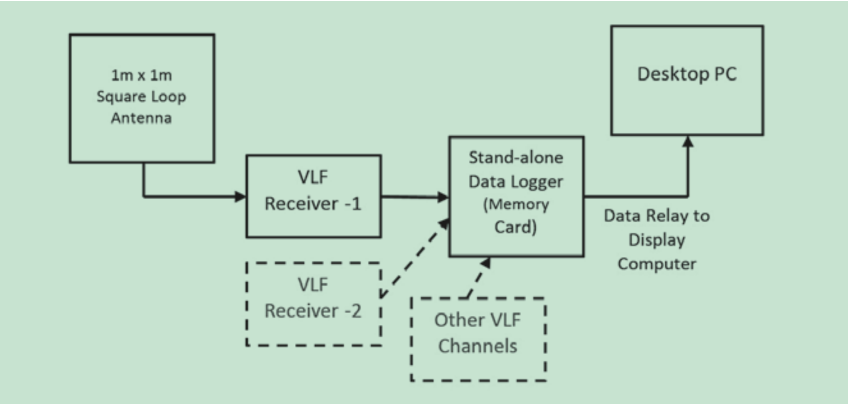

Fig. 2: The experimental setup depicting the antenna, VLF receivers, datalogger, and display computer

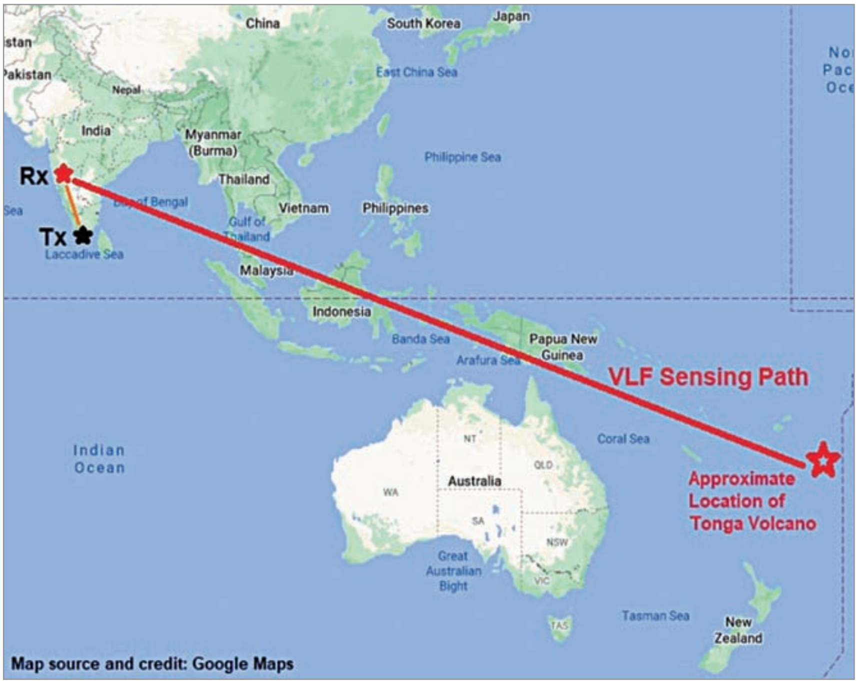

Fig. 3: The map showing VLF Tx, Rx, and the sensing zone extending from India to the South Pacific region

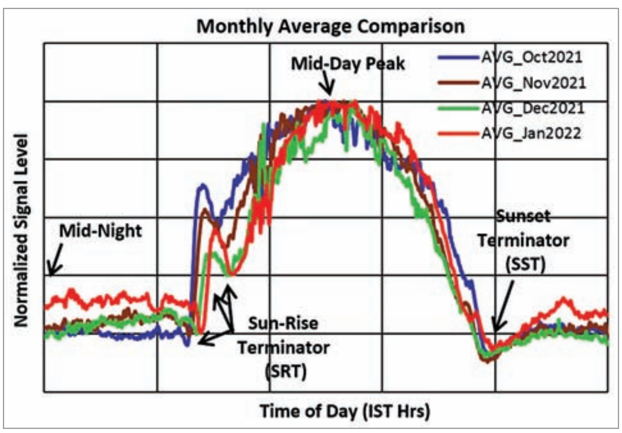

Fig. 4: Plots showing the nature of daily signal variations and comparison of monthly averages

are electromagnetic waves in the 3kHz to 30kHz frequency range with wavelengths ranging from 100km to 10km. Unlike the HF signals, however, the ionosphere acts as a reflector and helps in propagating these VLF signals in what is called Earth-Ionosphere-Waveguide-Mode of propagation.

It is the bottom of the ionosphere that is the D layer during day and E layer at night that forms one side of such a waveguide while the Earth's surface forms the other side of the waveguide. Fig. 1 illustrates this concept. Thus, the VLF signals propagate over longer distances virtually unattenuated. Therefore, these VLF signals are used for navigational applications.

Since these signals get reflected from the bottom of the ionosphere, monitoring the signal strength of VLF signals becomes a simple yet very effective and reliable tool for the remote sensing of the bottom of the ionosphere or simply the D layer or E layer. Any solar activity, such as solar bursts and coronal mass ejections (CMEs), have profound effect on D layer and enhancements in VLF signals are observed during such events.

Ionospheric disturbances in VLF propagation due to earthquakes, cyclones, and volcanoes are long been reported by scientists all over the world. However, such observations are reported only during or after such events.

The present work is therefore reporting for the first time observations of the pre-event signatures of a major volcanic eruption detected several weeks before the actual eruption: On 15th January 2022, an undersea volcano erupted in the small island country of Tonga in the South Pacific region, while the experimental setup running thousands of kilometres away in India was detecting abnormal changes since December 2021. This is a major scientific finding that reveals how and when the geophysical process of a volcanic eruption begins. It is a unique and first of its kind observation in this field.

The following sections describe